

ALI NIKKHAH

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SUMMARY

Union 10-page CV — all experiences, gap-free timeline with parallel annotations.

PROFESSIONAL EXPERIENCE (UNION)

Iran Credit Scoring Bureau

Tehran, Iran

ML Engineer — Self-Hosted Graph RAG Assistant (Concurrent)

Apr 2026 == Present

- Architected self-hosted Graph RAG assistant: Neo4j knowledge graph + vector retrieval fused into an agent that answers analyst queries over credit-scoring domain data with grounded citations.
- Ran self-hosted open-weight LLMs (vLLM/TGI) on GPU; profiled and optimised GPU runtime — KV-cache, batching, quantization (AWQ/GPTQ) — cutting per-token latency and GPU memory footprint.
- Implemented Neo4j tool-calls (Cypher generation + schema-aware retrieval) so the assistant executes graph traversals instead of hallucinating structured facts.
- Added hallucination guardrails — retrieval grounding, citation enforcement, deterministic fallback — for a regulated financial environment.
- *Stack*: Python, Neo4j, Cypher, Graph RAG, LangGraph, LangChain, FAISS, vLLM
- *Most recent Iran position; overlaps Sharif M.Eng. AI (Sep 2026) — full-time industry + part-time graduate study*

Turquoise Digital

Tehran, Iran

Senior Data Scientist & Data Engineer (Concurrent)

Sep 2025 == Mar 2026

- Centralized lake from raw CDC via PeerDB + Trino/Parquet on 5TB/day, ¡60s freshness; Airflow + PySpark DAGs cut memory 60%, governed by Great Expectations + Prometheus/Grafana.
- Built default-risk ensemble (XGBoost + time-series Transformers) — AUC 0.87, -22% defaults; 50+ concurrent mSPRT A/B tests with canary & auto-rollback, presented to C-suite.
- Shipped Whisper/wav2vec2.0 + LangGraph agentic chatbot — -78% ticket volume, 94% intent accuracy; deterministic fallback + policy guardrails for auditability.
- Drove portfolio allocation & risk engine (Sharpe +0.42) with streaming ClickHouse signals; LlamaIndex retrieval graph for ranking features.
- *Stack*: Python, PyTorch, Spark, Trino, PeerDB, Airflow, Great Expectations, Prometheus
- *Concurrent with Advanced Analytics Australia (Remote Contract) — 7-month overlap, split focus: Turquoise BNPL/risk/whisper vs Advanced edge CV*

Advanced Analytics Australia

Remote · Australia (Tehran-based)

Computer Vision / ML Platform Engineer (Edge) (Concurrent)

Sep 2025 == Apr 2026

- Deployed YOLO + ViT quantized (TensorRT/TF Lite) on Triton — 99.2% uptime ¡50ms p99, 3.5x speedup, exactly-once Kafka, 30 FPS over 100+ streams.
- Built edge MLOps with Triton model repository, zero-downtime rolling updates, automated rollback, Prometheus/Grafana per-site health.
- Cut false positives -35% via attention-guided NMS + SNR monitoring; broker architecture for 12 distributed sites.
- *Stack*: Python, OpenCV, YOLO, ViT, TensorRT, TFLite, Triton, Kafka
- *Parallel remote contract alongside Turquoise — edge CV track*

Sharif University — Robust & Interpretable ML Lab

Tehran, Iran

Research Collaborator — Explainable & Robust AI (Concurrent)

Jul 2025 == Sep 2025

- Benchmarked adversarial + natural shift suites; notebooks surfacing Captum/SHAP attributions, saliency, counterfactuals.
- Documented deploy best-practices for regulated industries with faculty.
- *Stack*: PyTorch, MLflow, Captum, SHAP
- *Summer research sprint concurrent with industry roles*

Trinity College Dublin — EmoDub Project

Dublin, Ireland

Machine Learning Researcher — Emotion-Aware Speech Translation (Concurrent)

May 2025 == Nov 2025

- Trained SER front-end (transformer + wav2vec2.0 + Whisper) for affect logits; cross-modal fusion transformer for emotion-consistent decoding.
- Curated emotion-annotated corpora (EmotionNet/EmoDB) with balanced cultural representation; evaluated BLEU, Emotion F1, MOS.
- Collab. with Prof. Siobhán Clarke; reproducible codebase <https://github.com/AliNikkhah2001/emodub-emotion-translation>.
- *Stack*: PyTorch, TensorFlow, Hugging Face, librosa, wav2vec2.0, Whisper, EmotionNet, EmoDB
- *Part-time remote research concurrent with Digikala + Turquoise + UBC*

Digikala

Tehran, Iran

AI Engineering R&D Lead (Concurrent)

Jan 2025 == Aug 2025

- Led team of 5 defining R&D roadmap + SLOs; shipped MCP with LangGraph cyclic graphs, zero-downtime hot-swapping, health-checked agent composition with deterministic fallback.
- Hybrid retrieval FAISS + Elasticsearch RRF + HyDE + KAG — +23% recall@10, -41% hallucination; multi-tier Redis caching, p99 j200ms.
- LLM serving on vLLM (PagedAttention) + Triton — 3.2x throughput, 60% cost reduction; 200 pods, MIG/HPA/VPA, Prometheus KV-cache dashboards.
- Partnered with infra/product establishing eval pipelines, A/B guardrails, canary rollouts.
- *Stack*: Python, LangGraph, LangChain, LlamaIndex, FAISS, Elasticsearch, ChromaDB, HyDE
- *Full-time lead; UBC/Trinity/Sharif were part-time remote (j10h/wk)*

Digikala

Tehran, Iran

AI Engineer (Concurrent)

Nov 2024 == Jan 2025

- Implemented LangGraph orchestration layers with health monitoring; hybrid retrieval with RRF + deterministic fallbacks for auditability.
- Embedded eval telemetry, memory-efficient indexing, response caching; load-tested to 5M queries.
- *Stack*: LangChain, LlamaIndex, FAISS, Elasticsearch, FastAPI
- *Promotion track — converted to R&D Lead Jan 2025*

University of British Columbia

Remote · Vancouver, Canada

Research Assistant — Vision-Language Medical Imaging (Ultrasound Report Generation) (Concurrent)

Feb 2024 == Jan 2025

- Designed ViT/BLIP encoders + T5 generator with contrastive/self-supervised alignment; curated DICOM, de-identified QA corpora.
- Deployed attention/saliency overlays for interpretability; benchmarked BLEU/ROUGE-L + clinical efficacy with radiologists; SOTA on in-house split.
- Preprint in prep: Automated DICOM-Compliant Ultrasound Report Generation via Contrastive Vision-Language Learning (2025).
- *Stack*: PyTorch, ViT, BLIP, T5, LLaVA, CLIP, OpenCV, MIMIC-CXR
- *Long-running part-time research; overlapped industry roles but capped at 10h/wk — primary commitment was Digikala/Turquoise*

Sharif University of Technology

Tehran, Iran

Teaching Assistant — NLP, Generative Models, Deep Learning (Concurrent)

Aug 2023 == Jan 2024

- Facilitated seminars, office hours (30-80 students), problem-set design/grading, exam design for DL/NLP.
- Developed lab material for Deep Learning, Analog Electronics, foundational ML; mentored undergrad capstones.
- *Stack*: PyTorch, TensorFlow, NumPy, scikit-learn
- *Graduate TA alongside L3S research — teaching load 6-8h/wk*

L3S Research Center — Leibniz University Hannover

Hannover, Germany

Research Assistant — Video Motion Classification (Concurrent)

Apr 2023 == Feb 2024

- Engineered optical-flow-centric datasets with temporal sampling/augmentation; improved domain generalization across lighting/texture/camera shift.
- Benchmarked 3D CNN + transformer hybrids vs baselines; analyzed attention maps; reduced overfitting, SOTA accuracy.
- *Stack*: PyTorch Lightning, OpenCV, PyAV, TensorBoard, scikit-learn, Multimodal Models
- *Part-time RA during B.Sc. final year — no conflict*

HomaCloud

Tehran, Iran

MLOps Engineer (Founding) (Concurrent)

Aug 2021 == Feb 2022

- Provisioned multi-node K8s via Terraform; Ray + MLflow training pipelines, TF Serving + FastAPI — j10 min releases, 99.9% deploy success, 15+ models.
- GitOps with GitHub Actions, canary, spot-instance optimization; Prometheus/Grafana + ELK observability; registry + lineage.
- *Stack*: Kubernetes, Docker, Ray, MLflow, Terraform, ArgoCD, FastAPI, Prometheus
- *During B.Sc. — clean sequential after Mapna internship*

HomaCloud

Tehran, Iran

MLOps Intern

May 2021 == Aug 2021

Sharif University of Technology

Tehran, Iran

Teaching Assistant — History (8 courses, 2021-2023)

Jan 2021 == Aug 2023

- Graded problem sets, ran tutorials, held office hours; supported 30-80 students per term.
- *Stack*: PyTorch, TensorFlow, NumPy

Mapna-MD2

Tehran, Iran

Full Stack Engineer

Jan 2021 == May 2021

- *Stack*: React, Node.js, PostgreSQL, Docker

EDUCATION

Sharif University of Technology

B.Sc. Electrical Engineering – Digital Systems *GPA: 3.65 overall, 3.91 major*

Tehran, Iran

Sept 2020 – Jul 2024